

Q1

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop. 10 marks

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.

Read 8 student names from the user.

Traverse the list using an Iterator.

Display each student name on a separate line.

Display the total number of students stored in the list.

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int count=0;
6         ArrayList<String> l=new ArrayList<>();
7         for(int i=0;i<8;i++){
8             l.add(sc.nextLine());
9         }
10        Iterator<String> it=l.iterator();
11        while(it.hasNext()){
12            System.out.println(it.next());
13            count++;
14        }
15        System.out.print("total number of students: "+count);
16        sc.close();
17    }
18 }
```

 Run Save

Input

```
sara
kitty
lisa
katty
christine
ava
mimi
julie
```

Output

```
sara
kitty
lisa
katty
christine
ava
```

Q2 Remove Even Numbers Using Iterator

10 marks

A company maintains a list of employee ID numbers. The management wants to remove all employees whose ID numbers are even, using an Iterator.

Write a Java program to:

Create an ArrayList<Integer>.

Read 10 employee ID numbers from the user.

Traverse the list using an Iterator.

Remove all even ID numbers using the remove() method of the Iterator.

Display the updated list containing only odd employee IDs.

main.java		Share	Settings	Save	Run	Output
<pre>1 import java.util.*; 2 public class main{ 3 public static void main(String[] args){ 4 Scanner sc=new Scanner(System.in); 5 ArrayList<Integer> l=new ArrayList<>(); 6 for(int i=0;i<10;i++){ 7 l.add(sc.nextInt()); 8 } 9 Iterator<Integer> it=l.iterator(); 10 while(it.hasNext()){ 11 Integer el=it.next(); 12 if((el%2==0)){ 13 it.remove(); 14 } 15 else{ 16 System.out.println("remaining employees: "+el); 17 } 18 } 19 sc.close(); 20 } 21 }</pre>						Status: Accepted Memory Used: 12436 byte Execution Time: 0.185 sec remaining employees: 13 remaining employees: 89 remaining employees: 45
						Input 12 13 14 34 22 89 76 56 45 10

Q3 Product Price Analysis Using Iterator

10 marks

A supermarket stores the prices of various products in an `ArrayList<Double>`. The manager wants to analyze the prices using an Iterator.

Write a Java program to:

- Create an `ArrayList<Double>`.
- Read the prices of 10 products from the user.
- Traverse the list using an Iterator.
- Calculate and display:
 - Total price of all products.
 - Average product price.
 - Highest product price.
- Display all product prices using the Iterator.

Your Code

```
1import java.util.*;
2public class main{
3    public static void main(String[] args){
4        Scanner sc=new Scanner(System.in);
5        ArrayList<Double> l=new ArrayList<>();
6        double sum=0;
7        double avg;
8        for(int i=0;i<10;i++){
9            l.add(sc.nextDouble());
10        }
11        Iterator<Double> it=l.iterator();
12        while(it.hasNext()){
13            double cp=it.next();
14            System.out.println(cp);
15            sum+=cp;
16        }
17        avg=sum/10;
18        System.out.println("total price of all products: "+sum);
19        System.out.println("average product price: "+avg);
20        l.sort(null);
21        System.out.println("highest product price: "+l.get(l.size()-1));
22        sc.close();
23    }
24}
```

Input

```
100
200.50
300.45
100.75
```

Output

```
50.32
total price of all products:
2325.4900000000002
average product price:
232.54900000000004
highest product price: 500.9
```